COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	7/29/2026	**DATE SUBMITTED:**	7/29/2026
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ACTIVITY NO. 2.1 **Control Structures (part 1)**

1. PROCEDURE OUTPUT

1. Read the student's score and determine if it's passed or failed

PASS.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main () {
    int score;

    cout << "Input your score: ";
    cin >> score;

    if (score >= 75) {
        cout << "You Passed!";
    }
    else {
        cout << "Sorry you failed :(";
    }
return 0;
}
```

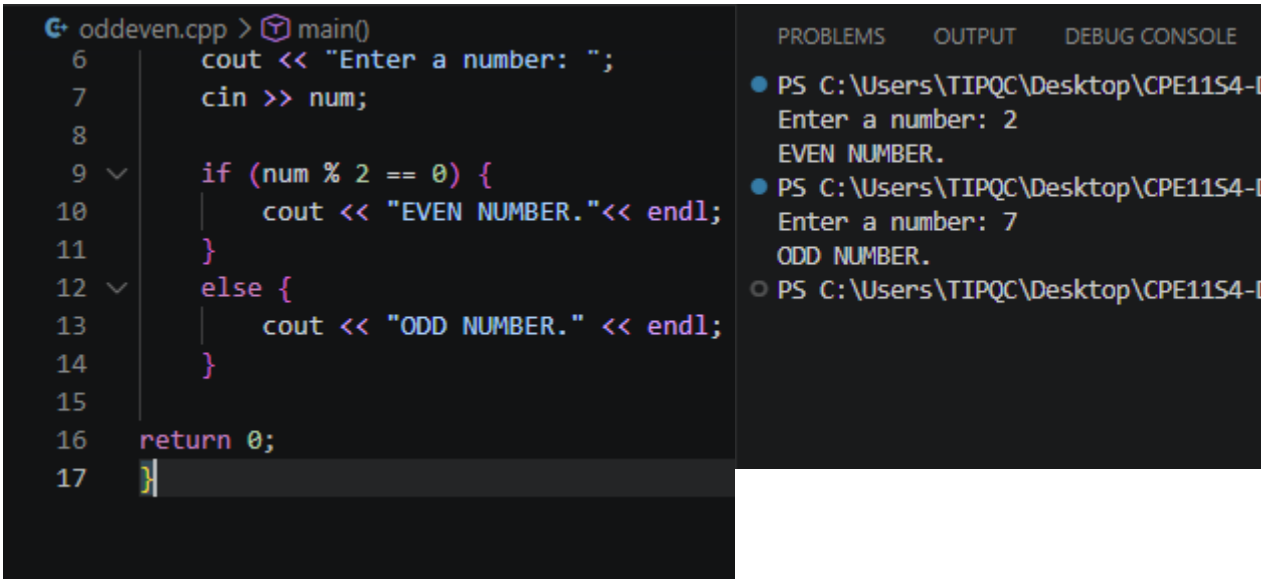
PROBLEMS OUTPUT DEBUG CONSOLE

```
PS C:\Users\TIPQC\Desktop\CPE11S4
Input your score: 75
You Passed!
PS C:\Users\TIPQC\Desktop\CPE11S4
Input your score: 57
Sorry you failed :(
PS C:\Users\TIPQC\Desktop\CPE11S4
```

This is a simple program that asks a user about a test score and determines if it is passing or failing. The code starts with the machine getting an input from the user using cin, which would then be processed into a block of decision that checks if the score is greater than or equal to 75, it is passed, otherwise the score will result on a failed mark.

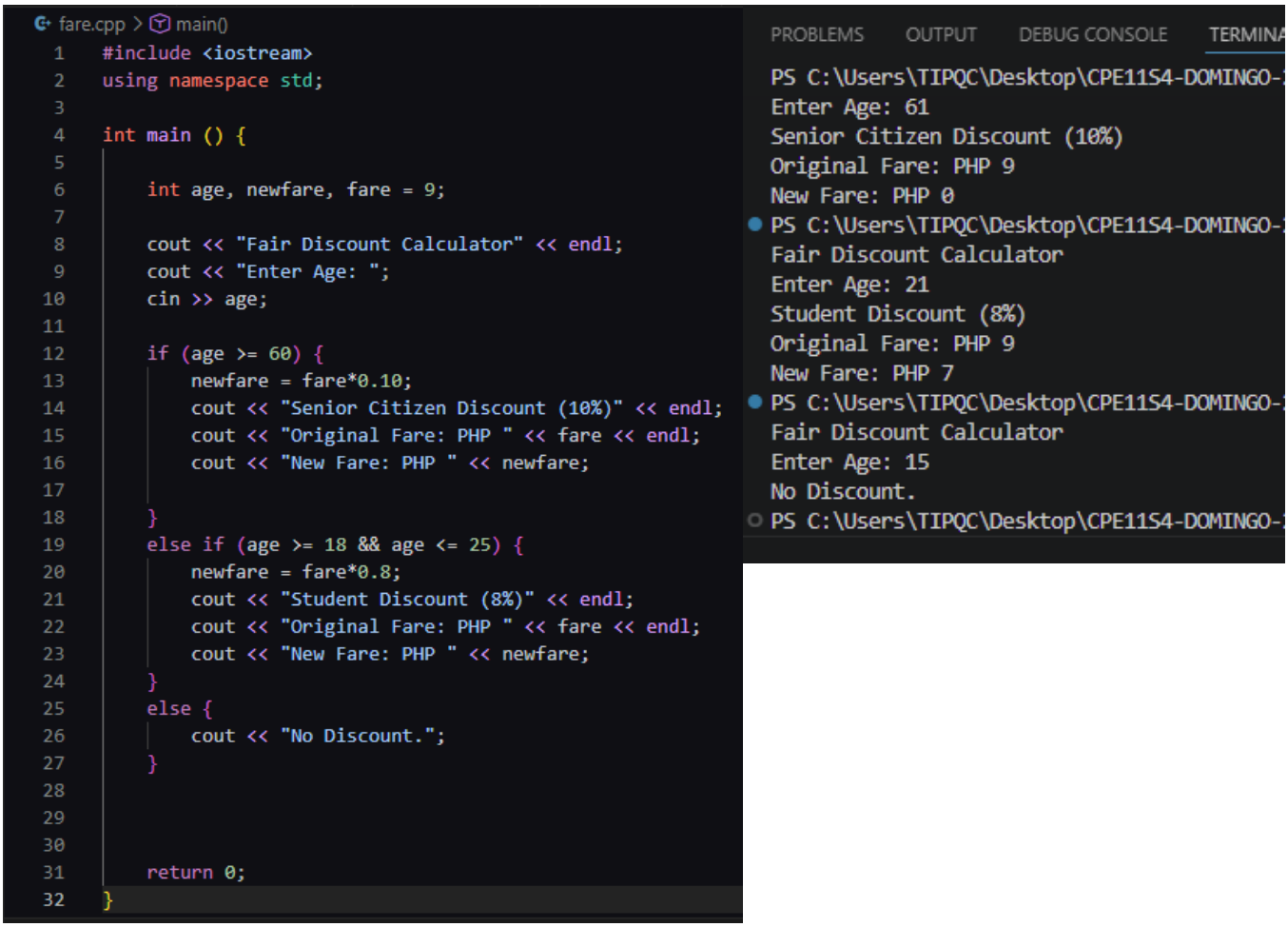
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Using conditional statements (if-else statements), write a program that asks a user for a number and prints out if it is an even or an odd number.



This is a program that takes a number from a user which then utilizes the modulo operator to determine if the number is even or odd. The main indication of an even or odd number will greatly depend on whether there is a remainder after dividing the provided number by the value of 2. If there is no remainder, it is even, if there is a remainder, then otherwise.

2. Using conditional statements, write a program that computes the fare discount based on the passenger's age. Passengers aged 60 and above are considered senior citizens and are entitled to a 10% fare discount. Passengers aged 18 to 25 are considered students and are entitled to an 8% fare discount. All other passengers will receive no discount. Prompt the user to enter their age. Assume that the minimum fare is ₱9.00. Display the original fare, the discount amount, and the final fare after the discount.



```cpp
#include <iostream>
using namespace std;

int main () {

    int age, newfare, fare = 9;

    cout << "Fair Discount Calculator" << endl;
    cout << "Enter Age: ";
    cin >> age;

    if (age >= 60) {
        newfare = fare*0.10;
        cout << "Senior Citizen Discount (10%)" << endl;
        cout << "Original Fare: PHP " << fare << endl;
        cout << "New Fare: PHP " << newfare;

    }
    else if (age >= 18 && age <= 25) {
        newfare = fare*0.8;
        cout << "Student Discount (8%)" << endl;
        cout << "Original Fare: PHP " << fare << endl;
        cout << "New Fare: PHP " << newfare;
    }
    else {
        cout << "No Discount.";
    }



    return 0;
}
```

```
PS C:\Users\TIPQC\Desktop\CPE11S4-DOMINGO-
Enter Age: 61
Senior Citizen Discount (10%)
Original Fare: PHP 9
New Fare: PHP 0
PS C:\Users\TIPQC\Desktop\CPE11S4-DOMINGO-
Fair Discount Calculator
Enter Age: 21
Student Discount (8%)
Original Fare: PHP 9
New Fare: PHP 7
PS C:\Users\TIPQC\Desktop\CPE11S4-DOMINGO-
Fair Discount Calculator
Enter Age: 15
No Discount.
PS C:\Users\TIPQC\Desktop\CPE11S4-DOMINGO-
```

The situation had a direction to calculate and display fare discounts, based on the passenger's age. Therefore, this program is designed exactly to do just that wherein it first asks the user the age before determining the corresponding fare discount and printing the new fare.

3. Case Study: Sentinel Controlled Repetition. Given the following pseudocode, create a program that will implement a sentinel controlled repetition. For example, you can use (-1) as the sentinel value. You can use Problem 1 as your reference.

Pseudocode:
Initialize total to zero
Initialize counter to zero
Input the first grade
While the user has not as yet entered the sentinel
Add this grade into the running total
Add one to the grade counter
Input the next grade (possibly the sentinel)
If the counter is not equal to zero
Set the average to the total divided by the counter
Print the average
else
Print “No grades were entered”

```
#include <iostream>
using namespace std;


int main() {

    double total = 0, counter = 0, grade, ave;

    cout << "Enter Grade (17 to terminate): ";
    cin >> grade;

    while (grade != 17) {
        total+=grade;
        counter++;

    cout << "Enter Grade (17 to terminate): ";
    cin >> grade;

    }
    if (counter != 0) {
        ave = total/counter;
        cout << "Average is: " << ave;

    }
    else {
        cout << "No grades were entered.";
    }


    return 0;
```

Output

```
Enter Grade (17 to terminate): 76
Enter Grade (17 to terminate): 98
Enter Grade (17 to terminate): 97
Enter Grade (17 to terminate): 89
Enter Grade (17 to terminate): 93
Enter Grade (17 to terminate): 17
Average is: 90.6

=== Code Execution Successful ===
```

This program uses the while loop to produce a continues loop of obtaining the grades where the sentinel, if inputed, will halt the loop and use the operation designated for getting the average of all the given grades by the users.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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